CHAPTER 9

Memory Hierarchy, Performance & Interfacing

Locality · AMAT · hierarchical vs simultaneous access · chip count · address decoding · interleaving

QUICK TRICK

AMAT = hit time + miss rate × miss penalty. Two levels: $H_1 + m_1(H_2 + m_2 \cdot P_{mm})$. Chips needed = **(total bits) ÷ (chip bits)**. Address bits = $\log_2$**(number of locations)**.

9.1 The hierarchy

● LEVEL 1 · BEGINNER

The problem. Fast memory is small and costly; big memory is slow and cheap. **The idea.** Stack memories from small/fast to big/slow so that most accesses hit the fast level, because programs show **locality**. **Analogy.** Your desk (cache), your bookshelf (main memory), the library (disk).

Level	Speed	Size	Cost / bit
Registers	fastest	tiny	highest
Cache (SRAM)	ns	KB – MB	high
Main memory (DRAM)	~50–100 ns	GB	medium
Secondary storage	ms	TB	lowest

- **Temporal locality:** a recently used item is likely to be used again soon (loops).
- **Spatial locality:** items near a used item are likely to be used (arrays, sequential code). Blocks (lines) exploit this.
- **SRAM:** flip-flops, fast, no refresh, costly (cache). **DRAM:** capacitors, needs refresh, dense and cheap (main memory).

9.2 Performance of a hierarchy

● LEVEL 2 · EXAM

```
AMAT = Hit time + Miss rate x Miss penalty
2-level: AMAT = H1 + m1 x (H2 + m2 x Pmem)          (m2 = local miss rate of L2)
Effective access time (hit ratio h, cache tc, memory tm):
   Simultaneous (parallel) access : T = h*tc + (1-h)*tm
   Hierarchical (sequential)      : T = h*tc + (1-h)*(tc + tm)
```

Example. Hit time 2 ns, miss rate 5%, penalty 100 ns: AMAT = 2 + 0.05 × 100 = **7 ns**. **Two level:** H_1 = 1 ns, m_1 = 10%, H_2 = 10 ns, m_2 = 20%, memory 100 ns: AMAT = 1 + 0.1 × (10 + 0.2 × 100) = **4 ns**.

Example. h = 0.9, t_c = 10 ns, t_m = 100 ns. Simultaneous: 9 + 10 = **19 ns**. Hierarchical: 9 + 0.1 × 110 = **20 ns**. **Speedup due to cache** = t_m / T = 100/19 ≈ 5.3.

COMMON MISTAKE

- Always read whether the question uses simultaneous or hierarchical access.

9.3 Memory interfacing (building memory from chips)

● LEVEL 2 · EXAM

- Memory size = (number of locations) × (word width). Chip = 2^a locations × d bits.
- **Chips needed** = (required locations ÷ chip locations) × (required width ÷ chip width).
- Arrange as **rows × columns**: columns widen the word, rows add locations. The high-order address bits go to a **decoder** that picks the row (chip select); the low-order bits go to every chip.

Example. Build a 64K × 16 memory from 16K × 8 chips. Chips = (64K/16K) × (16/8) = 4 × 2 = **8**: 4 rows × 2 columns. Address bits = 16. A_{13}...A_0 go to all chips; $A_{15}A_{14}$ drive a **2-to-4 decoder** selecting the row.

```
A15 A14 -> [2-to-4 decoder] -> CS of row0..row3
A13..A0 -> every chip          each row has 2 chips side by side: D15..D8 | D7..D0
```

Example. 16K × 8 memory from 4K × 8 chips: 4 chips, address 14 bits; $A_{13}A_{12}$ → 2-to-4 decoder; A_{11}...A_0 to each chip.

9.4 Address counting and interleaving

● LEVEL 2 · EXAM

- 64 KB byte-addressable → 16 address lines. If the machine is 32-bit word-addressable with 64 KB of storage: 16K words → 14 address lines.
- **Low-order interleaving:** with m banks ($m = 2^j$), the low j address bits choose the bank, so consecutive addresses fall in different banks and can be accessed in an overlapped way (good for sequential access).
- **High-order interleaving** = ordinary contiguous banks (good for adding capacity, no overlap).

9.5 Common mistakes

COMMON MISTAKE

- Mixing up bits and bytes in the chip-count formula.
- Using the global miss rate for L2 where the formula needs the **local** miss rate (or the other way round).
- Forgetting the 1 ns / 10 ns hit times when expanding a 2-level AMAT.
- Counting address lines for words when the question says byte-addressable.

GATE QUICK CHECK — Chapter 9

```
AMAT = Th + m*Tp       2-level: H1 + m1(H2 + m2*Pmm)
parallel: h*tc+(1-h)*tm     hierarchical: h*tc+(1-h)(tc+tm)
chips = (locs/chip_locs) x (width/chip_width)     address lines = log2(locs)
SRAM cache fast/costly | DRAM main memory refresh | low-order interleave -> consecutive banks
```

Practice questions

● LEVEL 1 · BEGINNER

Q1. Hit time 2 ns, miss rate 5%, miss penalty 100 ns. AMAT?

Q2. Address lines for a byte-addressable 64 KB memory.

Q3. Which memory is used for cache and which for main memory?

Q4. Name the two kinds of locality.

● LEVEL 2 · EXAM

Q5. h = 0.9, t_c = 10 ns, t_m = 100 ns. Effective time with simultaneous access and with hierarchical access?

Q6. Two-level cache: H_1 = 1 ns, m_1 = 10%, H_2 = 10 ns, local m_2 = 20%, memory 100 ns. AMAT?

Q7. How many 256K × 4 chips build a 1M × 8 memory? Arrangement?

Q8. Maximum miss rate so AMAT ≤ 3 ns with hit time 1 ns and penalty 50 ns?

● LEVEL 3 · GATE

Q9. Build 16K × 8 memory from 4K × 8 chips. Chips, address bits, and the decoder inputs?

Q10. Cache 20 ns, memory 200 ns, hit ratio 0.95 (simultaneous). Effective time and speedup due to the cache?

Q11. Low-order interleaving with 4 banks. Which bank holds address 13?

Answers

Q	Solution
1	2 + 0.05 × 100 = **7 ns**.
2	**16**.
3	Cache: **SRAM**. Main memory: **DRAM**.
4	**Temporal** and **spatial**.
5	Simultaneous: 9 + 10 = **19 ns**. Hierarchical: 9 + 0.1 × 110 = **20 ns**.
6	1 + 0.1 × (10 + 20) = **4 ns**.
7	(1M/256K) × (8/4) = 4 × 2 = **8 chips**: 4 rows × 2 columns.
8	1 + 50m ≤ 3 → m ≤ **0.04 (4%)**.
9	**4 chips**, 14 address bits; decoder: 2-to-4 on **A13, A12**; A11…A0 to every chip.
10	T = 19 + 10 = **29 ns**. Speedup = 200/29 ≈ **6.9**.
11	13 mod 4 = **bank 1**.

Next: Chapter 10 — Cache Memory Mapping.

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